The Conway–Monster Bridge: A Hybrid Computational Architecture via Moonshine (Rigor-First Revision v1.0)

Abstract

We formalize a hybrid architecture whose state spaces arise from the 2B centralizer of the Monster and the degree-2 piece of the moonshine module. The **classical substrate** is the irreducible C -module $W_{\text{class}} = V_{24} \otimes \Sigma_{4096}$ (dimension $98{,}304$), for $C := C_{\mathbb{M}}(2\text{B}) \cong 2^{1+24}\colon \text{Co}_1$. The **moonshine substrate** is the degree-2 Monster space $B := \mathbf{1} \oplus \mathbf{196{,}883} \subset V^{\natural}$ (with $V_1^{\natural} = 0$). We establish the multiplicity-one restriction

$$\mathbf{196{,}883} \downarrow_C \cong \ \mathbf{98{,}304} \oplus \mathbf{98{,}280} \oplus \mathbf{299},$$

identify the $\mathbf{299}$ inside $S^2(V_{24})$, construct C -equivariant idempotents splitting B , and prove a uniqueness (up to scalar) of the 2B-gate intertwiner $\Gamma \in \text{Hom}_C(B, W_{\text{class}})$. Verification protocols certify commutants and decompositions. No physics claims are made; “quantum” refers only to the moonshine/Monster substrate.

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1. Mathematical setting

Let $\mathbb{M}$ be the Monster. Fix a 2B involution $t \in \mathbb{M}$ and write

$$C := C_{\mathbb{M}}(t) \cong 2^{1+24}\colon \text{Co}_1, \qquad E \cong 2^{1+24}, \quad \text{Co}_1 \leq C/E.$$

Let V_{24} be the natural 24 -dim Co_1 -module and Σ_{4096} the unique spin irrep of E with central sign -1 . Set

$$W_{\text{class}} := V_{24} \otimes \Sigma_{4096} \quad (\dim = 98{,}304), \qquad B := \mathbf{1} \oplus \mathbf{196{,}883} \subset V_2^{\natural}.$$

We use the Griess form on B ; on W_{class} we use the tensor product of the natural Co_1 -form on V_{24} with an E -invariant form on Σ_{4096} (unique up to scalar).

Axioms. (A1) Central sign: the center of E acts by -1 on Σ_{4096} . (A2) W_{class} is irreducible as a C -module. (A3) Explosion: no Co_1 -stable $S \cong 24 \otimes 512$ is C -stable; adjoining any noncentral $x \in E$ forces closure to W_{class} . (A4) Degree-2 split: $\mathbf{196{,}883} \downarrow_C \cong \mathbf{98{,}304} \oplus \mathbf{98{,}280} \oplus \mathbf{299}$ with $\mathbf{299} \subset S^2(V_{24})$ and $\mathbf{1} \subset B$ the Griess axis.

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2. Core theorems

Theorem 2.1 (Multiplicity-one restriction). The restriction $\mathbf{196{,}883} \downarrow_C$ decomposes as $\mathbf{98{,}304} \oplus \mathbf{98{,}280} \oplus \mathbf{299}$, each with multiplicity one. Consequently, $\dim \text{Hom}_C(B, W_{\text{class}}) = 1$.

Sketch. Compare characters on C and use orthogonality; multiplicity-one follows from the absence of repeated constituents. Frobenius reciprocity gives the Hom dimension.

Theorem 2.2 (Explosion). For any noncentral $x \in E$, if $S \subset W_{\text{class}}$ is Co_1 -stable with $\dim S = 12{,}288 \cong 24 \otimes 512$, then $\mathrm{Cl}_{\langle \mathrm{Co}_1, x\rangle}(S) = W_{\text{class}}$.

Sketch. Extraspecial/Clifford theory excludes intermediate C -stable submodules between $24 \otimes 512$ and $24 \otimes 4096$.

Theorem 2.3 (Double commutant). $\dim \mathrm{Comm}_E(W_{\text{class}}) = 24^2 = 576$ and $\mathrm{Comm}_{E,\mathrm{Co}_1}(W_{\text{class}}) = \mathbb{C} \cdot I$.

Proof. $\mathrm{Comm}_E(V_{24} \otimes \Sigma_{4096}) \cong \mathrm{End}(V_{24})$ by Schur's lemma; adjoining Co_1 collapses to scalars by irreducibility of V_{24}.

Theorem 2.4 (Equivariant idempotents). There exist C -equivariant orthogonal idempotents $P_{\text{class}}, P_{\text{resid}}, P_{299}, P_{\mathbf{1}} \in \mathrm{End}(B)$ splitting B into the constituents $\mathbf{98{,}304}, \mathbf{98{,}280}, \mathbf{299}, \mathbf{1}$.

Construction. Character idempotents via group averaging; orthogonality is with respect to the Griess form.

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3. The bridge and its constraints

Bridge intertwiner. Define $\Gamma \in \mathrm{Hom}_C(B, W_{\text{class}})$ with $\mathrm{Im}\,\Gamma = W_{\text{class}}$. By Theorem 2.1, Γ is unique up to scalar.

Block projectors. Let

$$\Pi_\uparrow := P_{\text{resid}} + P_{299}, \qquad \Pi_\downarrow := P_{\text{class}}.$$

Then

$$\Pi_\downarrow \Pi_\uparrow = \Pi_\uparrow \Pi_\downarrow = 0, \quad P_i P_j = \delta_{ij} P_i, \quad \sum_i P_i = I_B.$$

> **Constraint (no equivariant round trip).** With multiplicity-one and non-isomorphic blocks, Schur's lemma implies $\mathrm{Hom}_C(W_{\text{class}}, W_{\text{resid}} \oplus W_{299}) = 0$. Thus any operational "up-map" (if used) is necessarily **non-equivariant** and lies outside the present math-only layer.

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4. Verification protocols (mathematical)

V1 (Dimension sanity). Verify

$$24 \cdot 4096 = 98{,}304, \quad 98{,}304 + 98{,}280 + 299 = 196{,}883, \quad \dim S^2(V_{24}) = \tfrac{24 \cdot 25}{2} = 300 = 1 + 299, \quad 24^2 = 576.$$

V2 (Idempotent certificates). Build P_i by averaging; certify $\|P_i^2 - P_i\| \le \tau$, $\|P_i P_j\| \le \tau$ for $i \neq j$ and $\|\sum_i P_i - I_B\| \le \tau$ (for a fixed operator norm).

V3 (Comm-dimension probe). Solve $[T, e] = 0$ for all $e \in E$ on W_{class} and check $\dim = 576$. Then add Co_1 -constraints and check collapse to scalars.

V4 (Explosion test). Choose Co_1 -stable $S \cong 24 \otimes 512$. For any noncentral $x \in E$, compute $\dim \mathrm{Cl}_{\langle \mathrm{Co}_1, x \rangle}(S)$ and certify $98{,}304$.

V5 (Degree-2 trace compatibility). On B , compute graded traces for classes $1, 2\mathrm{B}$ and compare leading coefficients with $J(\tau)$ and the 2B McKay–Thompson series (up to overall scaling of degree). Full J recovery requires higher degrees.

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5. Architectural reading (non-physical)

Classical substrate: the C -module W_{class} with structural control via E and Co_1 .

Moonshine substrate: the degree-2 space $B = \mathbf{1} \oplus \mathbf{196{,}883}$, decomposed by C -equivariant idempotents.

Bridge primitive: the unique-up-to-scale $\Gamma \in \mathrm{Hom}_C(B, W_{\text{class}})$. Any “up-map” from W_{class} to B that mixes blocks cannot be C -equivariant.

> **Remark on terminology.** We avoid physics interpretations; “quantum/moonshine substrate” denotes the Monster-module side only.

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6. References and provenance (mathematical)

Standard sources: Atlas of Finite Groups (Monster, 2B centralizer), Griess’ construction of $V^\natural$, Conway–Norton on McKay–Thompson series, and texts on extraspecial 2-groups and Clifford theory. (Add precise citations in implementation draft.)

Appendix A: Minimal statement of the bridge

There exists a C -equivariant orthogonal splitting $B = W_{\text{class}} \oplus W_{\text{resid}} \oplus W_{299} \oplus \mathbf{1}$ (multiplicity-one) and a nonzero $\Gamma \in \mathrm{Hom}_C(B, W_{\text{class}})$, unique up to scale. Any Co_1 -stable $24 \otimes 512$ slice explodes to W_{class} upon adjoining any noncentral element of E . Degree-2 graded traces on $\{1, 2\mathrm{B}\}$ match leading coefficients of $J(\tau)$ and the 2B McKay–Thompson series (up to degree scaling).